

Regime-Switching Asset Allocation via a POMDP

Type: New Application · 12 baselines, 5 diagnostic experiments

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Have you ever wondered...

Why does the “safe” 60/40 portfolio sometimes lose money when you most need it not to?

In **2022**, the textbook 60/40 stock-bond portfolio lost **~17%** as stocks AND bonds fell together.

Exactly the failure mode the 60/40 mix was sold as protecting against.

Why static allocation breaks

The 60/40 portfolio assumes a stable joint return distribution. Reality violates this:

- **Conditional heteroskedasticity** — calm decades broken by short crises (2008, 2020, 2022).
- **Stress correlation** — in crises, stocks and bonds sell off *together*. Diversification erodes exactly when it's most needed.

Question. Can a fully decision-theoretic regime overlay beat the static 60/40 after realistic transaction costs?

Why we're actually doing this

Money is the scoreboard. The *methods question* is the point:

- Test the implicit claim of every regime-switching paper.
- Build the policy from **explicit optimisation**, not heuristic.
- Treat the regime as a **hidden state** inferred by Bayes.
- Make the policy **falsifiable** (POMDP+QMDP).

*If we don't beat 60/40, we want to know **exactly which methodological choice** drives the failure.*

Before the math: five analogies

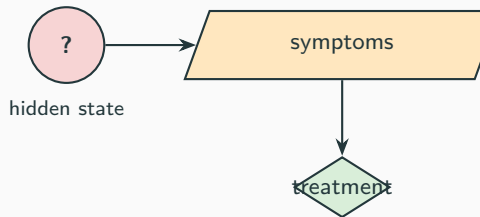
The next five slides explain every moving part of the project, *using no math*.

Doctor + symptoms	→	POMDP
Casino dealer with decks	→	HMM
Updating your guess	→	Bayes filter
Cards face-up	→	The underlying MDP
The dealer's whisper	→	QMDP

Analogy 1: the doctor with symptoms (= POMDP)

You are a doctor. Your patient is the market.

- Patient is *healthy* (bull) or *sick* (bear) — but you can't examine directly.
- You only see symptoms (VIX, term spread).
- Different treatments (portfolio weights) work better in different states.
- Goal: **read the symptoms, infer the state, prescribe.**



This IS a POMDP.

$s_t = \text{regime}$, $\mathbf{o}_t = \text{obs}$,

$\mathbf{a}_t = \text{weights}$.

Analogy 2: the dealer secretly switching decks (= HMM)

A poker dealer secretly swaps between two decks between rounds. You never see which deck. You only see the cards.

- **Transition matrix T** : how often the dealer switches. We estimate $\hat{T} \approx \begin{pmatrix} 0.96 & 0.04 \\ 0.06 & 0.94 \end{pmatrix}$ — sticky.
- **Emission $O(o | s)$** : the card distribution under each deck. Gaussian: $\mathcal{N}(\mu_s, \Sigma_s)$.
- **Baum–Welch (EM)**: from a long stream of cards, figure out each deck and the switch rate.

The Gaussian HMM is the formal model of this dealer.

Analogy 3: “which deck is in play *right now*?” (= Bayes filter)

After each card, you update your guess.

Predict: “The dealer might have switched.”

$$\hat{b}_t(s') = \sum_s T(s' | s) b_{t-1}(s).$$

Update: “A high card just came out — much more likely Deck A.”

$$b_t(s') \propto O(\mathbf{o}_t | s') \cdot \hat{b}_t(s').$$

One observation of VIX= 35 flips belief
from **70/30 bull** → **93/7 bear** in one step.

That responsiveness is the whole point of the filter.

Analogy 4: “what if the dealer dealt face-up?” (= MDP)

Suppose the dealer turned the decks face-up — you can see which deck is in use at every moment.

What’s the best action in each deck? Solve the Bellman equation:

$$V^*(s) = \max_{\mathbf{a}} \left\{ R(s, \mathbf{a}) + \lambda \sum_{s'} T(s' | s) V^*(s') \right\}$$

The answer is a table:

“In Deck A (bull regime):” do X.

“In Deck B (bear regime):” do Y.

This is the easier sub-problem. We solve it by value iteration AND policy iteration.

Analogy 5: “the dealer secretly whispers a probability” (= QMDP)

QMDP rule: act as if the dealer privately whispered “you’re in Deck s with probability $b(s)$,” then take the action that maximises your expected Q^* :

$$\pi_{\text{QMDP}}(\mathbf{b}) = \arg \max_{\mathbf{a}} \sum_s b(s) Q^*(s, \mathbf{a})$$

- Exact when $\mathbf{b}$ is a Dirac (fully observable limit).
- Cheap: reuses the MDP solver, no new optimisation per decision.
- Weakness: can’t motivate information-gathering — doesn’t bind here, regimes are exogenous.

Why POMDP, not a tree or fully-observable MDP

Naive alternative	Why it fails here
Decision tree	Explodes combinatorially with horizon
Influence diagram	Same blow-up, not natural for repeated rebalance
Fully-observable MDP	Requires knowing the regime — but it is <i>latent</i>
POMDP	Hidden state + noisy obs + sequential decision

In Powell (2019)'s 4-class taxonomy, QMDP is a **value function approximation** (linear lower envelope over the belief simplex) + a **one-step direct lookahead**.

The POMDP tuple, written once

$\mathcal{T} = \{0, 1, 2, \dots\}$, monthly epochs.

$\mathcal{S} = \{s_{\text{bull}}, s_{\text{bear}}\}$ latent macro regimes.

$\mathcal{A} =$
 $\{(0/100), (20/80), (40/60), (60/40), (80/20), (100/0)\}.$

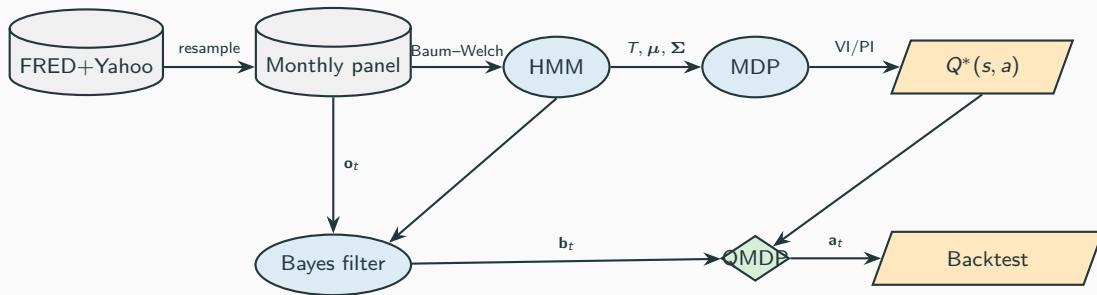
$T(s' | s)$ HMM transition matrix; **action-independent**.

$O(\mathbf{o} | s) = \mathcal{N}(\boldsymbol{\mu}_s, \boldsymbol{\Sigma}_s)$ on $\mathbf{o} = (\text{VIX}, \text{spread})$.

$R(s, \mathbf{a}) = \mathbb{E}[U(\mathbf{a}^\top \mathbf{r}) | s] - c \|\mathbf{a} - \mathbf{a}_{\text{prev}}\|_1$, CRRA U .

$\lambda = 0.95$ monthly discount.

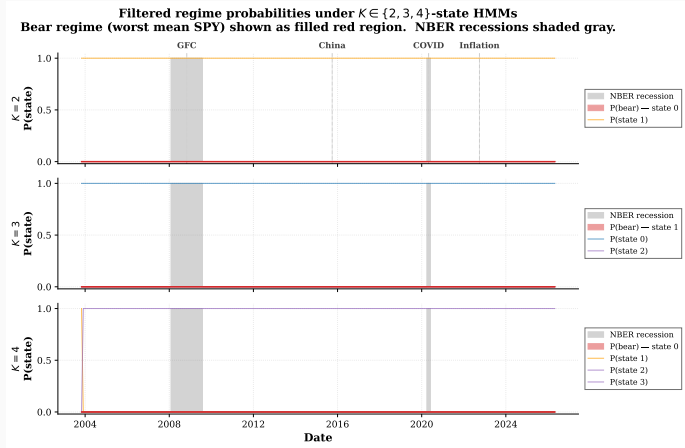
End-to-end pipeline



Top row: data path (raw $\rightarrow$ HMM $\rightarrow$ MDP $\rightarrow Q^*$).

Bottom row: decision-time loop (obs $\rightarrow$ belief $\rightarrow$ QMDP $\rightarrow$ weights).

HMM cleanly recovers known crises



Bear regime (red) tracks GFC 2008–12, China/oil 2015–16, COVID 2020, inflation 2022. NBER bars (gray) sit inside the recovered bear bands.

Fitted HMM parameters (2-state, 2003–2014 train)

Regime	Months	Mean SPY (%/mo)	Mean VIX
Bull (state 0)	87	+1.16	15.2
Bear (state 1)	48	−0.14	27.7

$$\hat{T} \approx \begin{pmatrix} 0.96 & 0.04 \\ 0.06 & 0.94 \end{pmatrix}$$

Both regimes are 95%+ sticky — the dealer rarely switches decks.

VI and PI agree exactly

The fully observable MDP satisfies

$$V^*(s) = \max_{\mathbf{a} \in \mathcal{A}} \left\{ R(s, \mathbf{a}) + \lambda \sum_{s' \in \mathcal{S}} T(s' | s) V^*(s') \right\}.$$

Value iteration: $V^{(n+1)} = LV^{(n)}$, stop at
 $\|\Delta V\|_\infty < 2.6 \times 10^{-6}$.

→ **125 iters.**

Policy iteration: solve $(I - \lambda P_\pi)V = r_\pi$,
greedy-improve.

→ **2 iters.**

Proposition: VI greedy policy = PI greedy policy.

Values agree within 2.6×10^{-6} . Sanity check passed.

QMDP rule: a worked numerical example

Belief after a high-VIX observation: $\mathbf{b} = (0.069, 0.931)$ (post-VIX-35 update).

At $\gamma=8$ (where the MDP policy actually differs across regimes):

Action	$Q^*(\mathbf{bull})$	$Q^*(\mathbf{bear})$	$\mathbf{b}^\top Q^*(\cdot, \mathbf{a})$
100/0 (all stocks)	0.32	0.10	0.115
60/40	0.24	0.20	0.203
40/60	0.20	0.22	0.219
0/100 (all bonds)	0.05	0.18	0.171

QMDP picks **40/60**: the bear-leaning belief steers toward bonds.

But at $\gamma=2$...

At our headline $\gamma=2$, the Q^* table looks different.
The column for “100/0 stocks” dominates **both** regimes.

QMDP always picks 100/0.

No matter what the belief.

This is the collapse.

We will trace exactly when and why this matters in the next act.

What if we benchmark against *everyone*?

Instead of comparing QMDP to just 60/40,
let's run it against **11 alternative strategies**
spanning the entire academic and practitioner consensus.

Static 60/40 · Equal weight · Inverse vol · Risk parity · Mean-variance · Faber 10-mo SMA ·
TS momentum · Vol-target · Myopic 12-mo · HMM-MV · Black–Litterman + HMM

Twelve-strategy horse race — the leaders

Strategy	CAGR	Vol	Sharpe	Sortino	Max DD	Calmar
Faber 10-mo SMA	17.24%	9.80%	1.68	3.57	−13.5%	1.28
Myopic 12-mo trend	15.34%	10.57%	1.41	2.70	−15.1%	1.01
TS momentum 12-mo	9.06%	8.25%	1.10	1.74	−22.0%	0.41
HMM-conditional MV	6.38%	5.91%	1.08	1.78	−18.4%	0.35
Black–Litterman + HMM	6.17%	6.65%	0.94	1.45	−18.0%	0.34
Inverse-vol	4.78%	5.24%	0.92	1.42	−16.8%	0.28

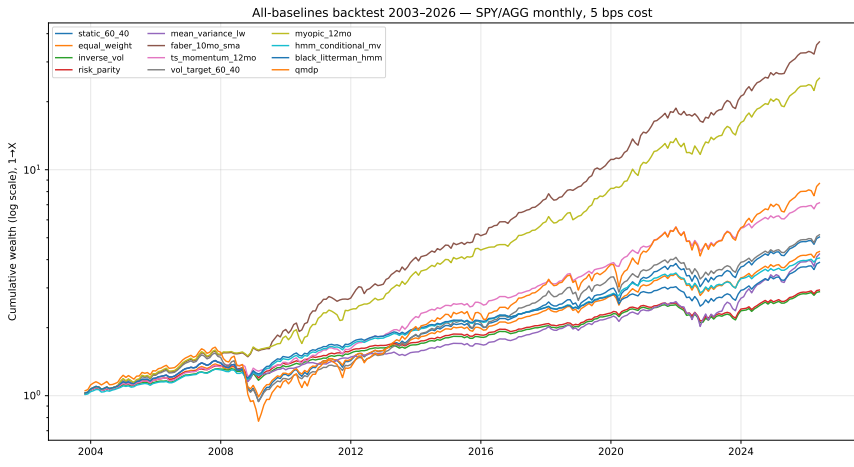
Faber's simple trend rule **dominates** on Sharpe, Sortino, and Calmar with the smallest drawdown among the leaders.

Twelve-strategy horse race — the laggards

Strategy	CAGR	Vol	Sharpe	Sortino	Max DD	Calmar
Risk parity (ERC)	4.87%	5.37%	0.91	1.41	−16.8%	0.29
Equal weight 50/50	6.69%	8.12%	0.84	1.26	−28.6%	0.23
Mean-variance + Ledoit-Wolf	6.58%	8.20%	0.82	1.32	−22.5%	0.29
Static 60/40 (benchmark)	7.40%	9.35%	0.81	1.21	−34.2%	0.22
Vol-target 60/40	7.50%	10.14%	0.77	1.12	−39.1%	0.19
QMDP ($\gamma=2$)	10.01%	14.64%	0.73	1.07	−53.0%	0.19

QMDP finishes dead last on Sharpe with the deepest drawdown of the twelve. The two HMM-aware non-QMDP baselines (HMM-MV, BL+HMM) *above* 60/40 — the HMM signal IS informative.

Equity curves: \$1 invested 2003-10-31 (log scale)



Faber (top) dominates. QMDP (bottom-right) has the highest equity-only CAGR but the worst Sharpe and worst drawdown.

Why Faber wins: “surf the wave”

Trend-following doesn't try to predict the next wave.

It just rides the one already breaking.

Hold an asset only if its price is above its 10-month moving average.

No regime model. No Bayesian filter. No MDP. Just trend.

Sharpe 1.68, max drawdown –13.5%, Calmar 1.28.

Consistent with Hurst/Ooi/Pedersen 2017 “A Century of Evidence on Trend-Following.”

Why QMDP collapsed: γ never separates regimes

Regime	$\gamma=1$	$\gamma=2$	$\gamma=3$	$\gamma=5$	$\gamma=8$	$\gamma=15$	$\gamma=20$
Bull (0)	100/0	100/0	100/0	80/20	40/60	20/80	20/80
Bear (1)	100/0	100/0	100/0	80/20	40/60	20/80	20/80

- **Within columns:** bull and bear give the *same* action. Regime label never moves the policy.
- **Across columns:** higher γ shifts the policy bondward across-the-board.

Punchline: the low- γ gambler's analogy

A gambler with *low risk aversion*
bets the same way *regardless* of suspicion.

At $\gamma=2$, the long-run discounted continuation value of the high-EV action dominates the single-period downside — even when the belief is 93% bear.

Only $\gamma \geq 8$ lets suspicion shift the policy.

The Sharpe gain from raising γ is a vol-reduction effect, not a regime-aware effect.

Wait — what if we changed *one* thing?

We ran three diagnostic experiments.
Each one reverses the headline *in isolation*.

1. Add stress observations → regime policy flips
2. Walk-forward refit → Sharpe lifts to 1.08
3. Raise γ to 8 → QMDP beats static

The negative headline is a diagnostic of methodology, not fundamentals.

Unlock 1: richer observations *fix* the policy

Cohort	Features	Bull	Bear	Differs?
1. Baseline	VIX, T10Y3M	100/0	100/0	X
2. +Yield curve	+T10Y2Y	100/0	100/0	X
3. +Stress	+NFCI, +STLFSI4	100/0	0/100	✓ full flip
4. +Macro	+UMCSENT, +ICSA	100/0	0/100	✓
5. Kitchen sink	8 channels	100/0	60/40	✓

Same $\gamma=2$ as headline. We just changed which observations feed the HMM.

Punchline: “more thermometers, better diagnosis”

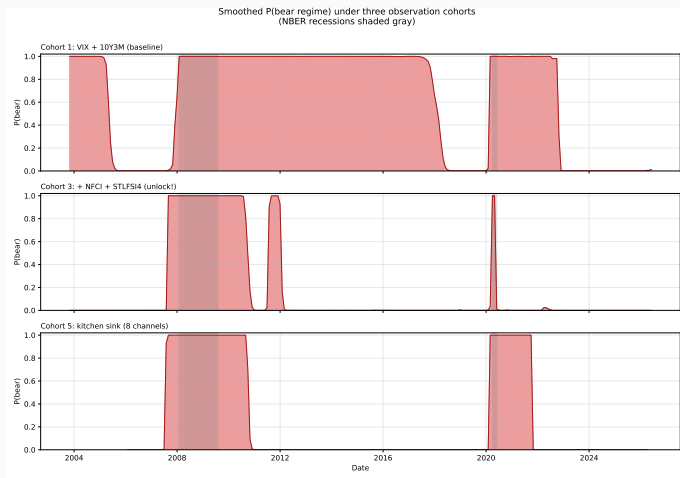
Two thermometers (VIX, term spread) measure the same axis: equity fear.

Adding **NFCI** (Chicago Fed) + **STLFSI4** (St. Louis Fed) is like adding a blood test and an X-ray.

With independent stress signals, the HMM differentiates a bear regime where conditional bond return meaningfully beats conditional stock return.

The headline collapse is an observation-channel problem, not a QMDP problem.

Visual confirmation: cohort 3 vs cohort 1



Cohort 1 (top, baseline): noisy. Cohort 3 (middle, +NFCI +STLFSI4): sharp high-confidence bear spikes cleanly bracket every NBER recession. Cohort 5 (bottom): over-fits.

Unlock 2: walk-forward refit lifts Sharpe to 1.08

Variant	CAGR	Sharpe	Max DD	Calmar
Static 60/40 (benchmark)	7.40%	0.81	−34.2%	0.22
QMDP fixed (report baseline)	9.87%	0.96	−34.2%	0.29
QMDP annual refit (5y win)	9.11%	0.85	−25.1%	0.36
QMDP quarterly refit	8.56%	0.83	−23.4%	0.37
QMDP expanding window	9.81%	1.08	−16.5%	0.60
HMM-MV expanding window	5.32%	0.95	−16.8%	0.32

Following Nystrup–Madsen–Lindström (2018).

Punchline: “re-read the casino’s recent hands”

A fixed HMM is like a player who learned the casino’s edges in 2003 and never updated.

Expanding-window walk-forward refit:

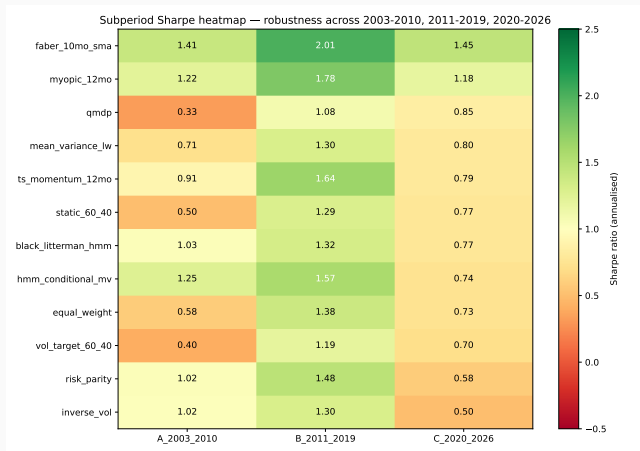
Sharpe 0.81 → **1.08** (+33%)

Max DD −34% → −**16.5%** (more than halves)

Calmar 0.22 → **0.60** (nearly triples)

The fixed-window HMM was a methodology choice masking real performance.

Subperiod robustness: Faber wins everywhere



Period A (2003–10): GFC. Period B (2011–19): post-GFC ZIRP. Period C (2020–26): COVID + inflation.

Punchline: QMDP fails worst exactly where it should help

QMDP's worst column is **Period A (GFC era)** with Sharpe **0.33**.

Exactly the period where a regime-aware model should help most.

In the same Period A, HMM-aware *non-QMDP* baselines (HMM-MV and BL+HMM) achieve Sharpe **1.03–1.25**.

The HMM signal was there. QMDP's $\gamma=2$ projection destroyed it.

The honest verdict: what we claim vs what we don't

We DO claim	We do NOT claim
POMDP is the right <i>minimum</i> tool for this decision problem.	POMDP asset allocation beats simple trend-following.
The headline negative finding is a diagnostic of methodology, not fundamentals.	The original QMDP at $\gamma=2$ is a useful production policy.
The HMM signal is informative (HMM-MV, BL+HMM beat QMDP).	Our walk-forward QMDP is the new state-of-the-art.
Three components separately reverse the negative finding.	Our gains net of transaction costs matter to a real allocator.

Three extensions in priority order

1. **CVaR-penalised reward** — replace CRRA with $R = \mathbb{E}[r] - \kappa \text{CVaR}_\alpha[r]$ to explicitly penalise bear-regime tail risk. One-line change.
2. **Point-based value iteration (PBVI)** — tighter α -vector bound on the 1-D belief simplex. Cheap here.
3. **Off-policy MC + Double Q-learning** (Precup 2000 + Van Hasselt 2010) — a model-free baseline that doesn't assume the QMDP transition matrix.

Public repository (one-clone, no API key):

`github.com/takakhoo/ENGS177_Final_Project`

- 14 raw data CSVs (~ 745 KB) + 3 pickled HMMs — *all committed*
- 4 source modules (hmm, mdp, qmdp, baselines)
- 12 experiment scripts (00...12, no 05)
- 11-page class report + 38-page extended technical report
- This 36-frame deck + 7 docs (project story, intuition primer, surveys)

Questions?